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Towards a Part-of-Speech (PoS) gram approach to academic writing: A case study of research introductions in different disciplines

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Abstract

This study innovatively applies the Part-of-Speech-gram (PoS-gram) procedure to the examination of language patterning and variability in a largely conventionalized part-genre (i.e., research introductions). Based on 400 article introductions from computer engineering (CE) and cognitive linguistics (CL), the study has identified key PoS-grams and their associated lexico-grammatical frames, using the written academic component of British National Corpus as the reference corpus. The analysis reveals key PoS-grams shared in CE and CL introductions, e.g., those associated with the step ''purposive announcement'', as well as the discipline-specific ones such as the PoS-gram for structure-outlining only found in CE introductions. Compared to various forms of multi-word sequences like n-grams, the PoS-gram has the unique strength of grouping phraseologies with similar or identical structure and discursive functions and yet either recurrent or varying lexical choices under the co-selected grammatical categories. The advantage enriches analyses and helps yield pedagogically useful findings, in that patterning and variability is revealed not only in the overall function, structure and composition of PoS-grams but in such aspects of their recurrent or diversified tokens. This study illustrates the innovative application of corpus-based PoS-gram procedure to academic genres, which may inspire a promising new line of inquiry and the current genre pedagogy.
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1. Introduction

The usefulness and effectiveness of corpus linguistics approaches to patterned language use in specialized genres have been increasingly recognized (Breeze, 2019; Golparvar and Barabadi, 2020; Güngör and Uysal, 2020), as they are able to help profiling what is featured by such (part-/para-) genres of a conventionalized nature. Among concepts related to corpus-approaches to language patterning, *part-of-speech-gram* (hereafter PoS-gram) is an important notion (Stubbs, 2007), which has however been largely overlooked in EAP research.

A *PoS-gram*, as defined by Stubbs (2007, p. 91), is ''a string of part-of-speech categories'', ''the tokens of which are strings of words that have been annotated with these PoS tags'' (Pinna and Brett, 2018, p. 107). Stubbs (2007) considered

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it a type of ''routine phraseology'', in addition to *n-grams* and *phrase-frames*. *N-grams* refer to recurrent contiguous multi-word sequences (e.g., *as a result of*) and can also be called *lexical bundles* (Biber et al., 1999), *clusters* (Hyland, 2008), *recurrent word-combinations* (Altenberg, 1998), *multi-word sequences* (Butler, 2003) and *chains* (Stubbs and Barth, 2003). *Phrase-frames* denote ''an n-gram with one variable slot'' (Stubbs, 2007, p. 90). Based on a powerful phraseology data-base developed by Fletcher (2003/2004), Stubbs identified the most frequent PoS-grams of length 5 in the British National Corpus (BNC) and described their example tokens, mostly being ''parts of nominal and prepositional phrases, which express spatial, chronological and logical relations'' (Stubbs, 2007, p. 94).

While these findings are intriguing, PoS-grams in our understanding may not be a type of phraseology (Stubbs, 2007), as phraseology is generally defined in corpus linguistics research as ''the recurrent co-occurrence of words'' (Clear, 1993, p. 277). Yet, the compositional unit of a PoS-gram is a PoS category (grammatical category) rather than a word form. Accordingly, we only treat it as a phraseology-related concept, since the exponents of each PoS-gram may be potential phraseology (Section 3) and the identification of it can be an effective way to extract recurrent phraseologies and patterns (Pinna and Brett, 2018).

In corpus-based EAP research, PoS-grams have received extremely scant attention. This is in stark contrast to the abundance of research on *multi-word sequences* (*phraseology*). It might be attributed to the far-reaching impact of the Sinclairian tradition of corpus linguistics research (Sinclair, 1991), which does not favor corpus annotation but ''takes the word-form as the focal point in the presentation of data'' (Hunston and Francis, 2000, p. 18). While the past three decades have witnessed the focus of corpus linguistics research on lexical and phraseological issues such as word frequency, keywords and various forms of the recurrent co-occurrence of words (i.e., *phraseology*), ''a renewal of interest in grammatical issues'' has been noted (e.g., Biber et al., 2020; Larsson and Kaatari, 2020) (Breeze, 2019, p. 80). The under-investigated PoS tag sequences and individual PoS categories have started to captivate scholarly attention for studying specialized genres such as the fiction and newspaper genres (Thompson and Sealey, 2007; Brett and Pinna, 2015; Pinna and Brett, 2018) and the legal register (Breeze, 2019). Yet, hardly any systematic research exists on PoS-grams in research articles (RAs) and their sections to examine their language patterning for both pedagogical and research insights.

Specifically, Thompson and Sealey (2007) compare the linguistic properties of the fictional prose written for children and that for adults and of newspaper texts through deploying word/PoS tag frequency analyses, word and PoS sequence analyses, and semantic analyses. The PoS-gram analysis, as they maintain, ''makes possible a novel perception of the frequency of the complex prepositional phrase'' (Thompson and Sealey, 2007, p. 21). The presence of this complicated prepositional phrase *(prep art NN1 of art NN1*[1]) together with the other two PoS-grams (*art adj NN1 prep art NN1*[2]) and (*art NN1 prep art adj NN1*[3]) is shared in all the three types of texts. Nevertheless, the newspaper texts still contrast remarkably with the two fiction corpora in the other linguistic analyses such as the multi-word sequence analysis. For instance, the news genre comprises sequences denoting causality (*in the wake of*) and those with a higher level of metaphoricity (*at the heart of the controversy*), whereas the two fiction corpora contain overwhelmingly the expressions of location, direction and temporality (*at the top of the hill; to the end of the lane; at the end of the day*) (p. 14). Noticeably, the sets of phraseologies characterizing genres as illustrated above, such as those referring to location, direction and temporality in the fiction genre, are analogous in forms and may be subsumed under the same PoS-grams with a higher level of generality to indicate recurrent grammatical structure and syntactic patterns.

Both Brett and Pinna (2015) and Pinna and Brett (2018) have validated the effectiveness of PoS-gram analyses for extracting phraseologies and patterns for the newspaper register. Brett and Pinna have corroborated the distinctive use of the inflected superlative adjective in tourism writing based on their PoS-gram analysis of 0.45 M token of travel journalism texts collected from the BBC web, with the 100 M token BNC as the reference corpus. Their analysis has identified a very limited number of highly frequent constructions featuring such inflectional superlatives. Given rather low variation in lexical choices within these constructions, they reflected on the utility of PoS-grams in capturing sets of phraseologies with similar functions and forms, and suggested their unique strengths and suitability for analyzing specialized genres (as compared to n-gram analysis).

Pinna and Brett (2018) undertook a PoS-gram analysis of 10 sub-registers of the newspaper *The Guardian*, with BNC as the reference corpus. Their analysis has clearly demonstrated the usefulness of the PoS-gram procedure for extracting *loose formulae* (Pinna and Brett, 2018, p. 121). *Loose formulae* refer to schemas with particular discursive functions, derived from *semi-prepackaged phrases* (Francis, 1993; Philip, 2008, 2011) or phraseologies grouped under statistically significant PoS-grams, with their components consistently playing certain semantic and functional roles. In addition to extracting such formulae which reveal patterned language features of specialized sub-registers, their PoS-gram analysis

[1] Preposition+Article+Singular noun+*of*+Article+Singular noun.
[2] Article+Adjective+Singular noun+Preposition+Article+Singular noun.
[3] Article+Singular noun+Preposition+Article+Adjective+Singular noun.

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shows linguistic variations and commonalities across them. To illustrate, the statistically significant PoS-grams AT0 AJ0 NN1 PRP AT0 NN1[4] (Example token: *a familiar figure in the town*), AJ0 NN1 PRP AT0 AJ0 NN1[5] (Example token: *small town on the Dutch coast*), AT0 AJ0 NN1 PRP AT0 AJ0[6] (Example token: *a perfect accompaniment to a cold*) and PRP AT0 AJ0 NN1 PRP AT0[7] (Example token: *on a rocky shelf above a*) have been shared by Travel and Obituaries. As seen above, the recurrent patterns in these two sub-registers concern the description of entities, whereas another sub-register Crime features constructions related to ''entities undergoing processes'' (Pinna and Brett, 2018, p. 116), as evidenced by one of its top-ranking PoS-grams (NN1 VBD VVN PRP AT0 NN1,[8] Example token: *body was found in the garage*).

Different from the above three studies focusing on PoS sequences, Breeze (2019) investigates key individual PoS categories across the four genre families in business law (viz., academic texts, case law, legal documents and legislation), with the BNC as the reference corpus. Her analysis has revealed that NNSZ (plural possessive noun) is key in all four genre families, suggesting it as the eminent grammatical feature of the legal register. However, grammatical specificity of these four genres have also been recorded: documents and legislation show ''unusual patterns of cohesion and modality'', whereas legal academic writing and case law ''follow patterns comparable to argumentative texts'' (Breeze, 2019, p. 79), featured with the frequent use of relative clauses and present tenses. Albeit these enlightening findings, Breeze recommends exploring PoS sequences, or combinations of PoS with specific lexical items for further research to increase our knowledge on language use in legal genres.

All the existing few PoS-gram studies have confirmed considerable potential and effectiveness of the PoS-gram procedure for specialized genre research. Yet, almost no study has hitherto systematically applied the PoS-gram analysis to RAs and their part-genres such as the largely conventionalized Introduction to be studied in this paper (Swales, 1990, 2004).

Previous introduction studies using genre and/or corpus approaches (e.g., Gledhill, 2000; Lu et al., 2018; Swales, 1990, 2004) have repeatedly verified the patterned structure of language use in this section, suggesting its suitability for a PoS-gram analysis. In terms of its rhetorical structure, the well-established Creating a Research Space (CARS) model and its revised version (Swales, 1990, 2004) have suggested three obligatory moves (viz., Move 1 *Establishing a territory*, Move 2 *Establishing a niche* and Move 3 *Occupying a niche/Presenting the present work*), under which steps of varying importance are subsumed (e.g., the only obligatory step of *outlining the purpose* in Move 3). Numerous follow-up genre analyses have corroborated the validity of the CARS model and its revised version, with some variations noted across disciplines, genres and languages (e.g., Anthony, 1999; Kanoksilapatham, 2012; Lin, 2014; Sheldon, 2011).

Concerning its language patterning, most scholarly attention has so far been devoted to different forms and variants of phraseology, most notably Bondi (2010) on *semantic sequences*, Cortes (2013) on *lexical bundles*, Gledhill (2000) on *collocations*, and Lu et al. (2018) on *phrase-frames*. Gledhill (2000) examines the collocations of salient grammatical words and their discourse functions in pharmaceutical sciences RA introductions, from which semantic patterns associated with the salient grammatical words are derived, which resemble *semantic sequences*[9] (Hunston, 2008). Along this line, Bondi (2010, p. 99) has discovered outline introductory formulae and ''framework semantic sequences'' associated with meta-discursive practices in economics RA introductions. Further, Cortes (2013, p. 41) has demonstrated the close connections between bundles and moves/steps of introductions and identified lexical bundles acting as *triggers* of moves and steps (e.g., *the purpose of the present study, the objective of this study*) and as *complements* (''used in the second part of the clauses or phrases identified as belonging to a particular step'', e.g., *in the sense that*). The trigger-type bundles together with many pedagogically useful phrase-frames identified by Lu et al. (2018) from 600 social science RA introductions (e.g., the verb-based frames ''*we find [little, no, strong, suggestive, weak] evidence that*'' and other-content-word frame ''*in the present study we [investigated, examine(d), focus, test(ed)]*''[10]) have all suggested recurrent patterns and phraseologies with similar forms and functions (and yet variability in lexical choices in certain PoS position). The bundles in isolation together with the phrase frames and their variants could be grouped under corresponding PoS-grams for a better view of overall (syntactic) patterns and internal variability.

Compared to n-grams that focus on ''identical, rather than very similar, strings'' (Pinna and Brett, 2018, p. 108) and phrase-frames with only one open slot for lexical variants, PoS-grams may have the strengths of better uncovering variability and grouping patterns. This is because in every slot of a PoS-gram, ''any word can occur as long as it belongs to

[4] Article+Adjective+Singular noun+Preposition+Article+Singular noun.
[5] Adjective+Singular noun+Preposition+Article+Adjective+Singular noun.
[6] Article+Adjective+Singular noun+Preposition+Article+Adjective.
[7] Preposition+Article+Adjective+Singular noun+Preposition+Article.
[8] Singular noun+*Was/Were*+Past participle of lexical verb+Preposition+Article+Singular noun.
[9] *Semantic sequences*, as defined by Hunston (2008: 271), are ''recurring sequences of words and phrases that may be very diverse in form and which are therefore more usefully characterized as sequences of **meaning elements** rather than formal sequences''.
[10] The words in square brackets, according to Lu et al. (2018: 81), suggest ''variants that fill the open slot in each frame''.

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Table 1
The sizes and composition of CCL and CCE.

Corpora	No. of texts	Tokens	Types
CCL	200	272,657	15,769
CCE	200	179,990	15,096

the PoS category of that particular position'' (Brett and Pinna, 2015, p. 52). The potential advantages of PoS-grams over multi-word sequences have been neatly summed up by Brett and Pinna (2015).

The quantitative analysis of strings of PoS categories and their relating tokens casts a looser net over a wider area, allowing us to discover widespread and characteristic patterns that fly below the statistical radar of more traditional and stricter forms of analysis such as n-grams, which can only reveal identity and not similarity. (p. 57).

Given such unique strengths of PoS-grams and the need to bridge the gap in EAP research, this study applies key PoS-gram analyses to the largely conventionalized article introductions from two contrasting disciplines (viz., cognitive linguistics (CL) and computer engineering (CE)), with the written academic component of BNC as the reference corpus. The research aims are three-fold: (1) methodologically, to examine the feasibility and usefulness of the PoS-gram analysis to language patterning in a conventionalized academic part-genre such as the Introduction section; (2) to generate patterns and phraseologies represented by the key PoS-grams characterizing this part-genre and to study the relevant cross-disciplinary commonalities and differences; and (3) to examine the links, if any, between the key PoS-grams specific to this part-genre and its functional moves and steps and the related cross-disciplinary commonalities and variations. It illustrates an innovative methodology of combining corpus-based PoS-gram analysis with EAP genre research, which is expected to inspire a promising new line of inquiry and shed new lights on the current EAP pedagogy.

2. Corpus compilation and analysis

2.1. Corpus compilation

To fulfill the research purposes, we have used AntCorGen 1.1.2 (Anthony, 2019) to collect 400 RA introductions with 200 each from the two disciplines (viz., CL and CE) to compile the two corpora, i.e., Corpus of Cognitive Linguistics introductions (CCL) and Corpus of Computer Engineering introductions (CCE). AntCorGen 1.1.2, as introduced by its developer Lawrence Anthony from Waseda University in its help file, is a reliable free computer software for corpus generation. One of its important relevant functions is that it can help researchers to download different sections (e.g., title, introduction) of journal articles from the PLOS ONE[11] research database based on different subject category (i.e., disciplines and sub-disciplines). By adopting this function, we have gathered all the introductions needed and their corresponding PDF full-texts (in case of the need to reference the whole research) from the areas of CL (under the category of ''Social Sciences''—sub-category ''Linguistics'') and CE (under the category of ''Engineering and Technology''—sub-category ''Electronics Engineering''). After downloading all texts needed, we have checked through them to ensure that all the introductions gathered are from empirical RAs, rather than from other categories of RAs such as theoretical or review papers.

Table 1 presents descriptive details of the two compiled corpora. As can be seen, their sizes are not very large, due to the restricted lengths of journal articles themselves and their introductions. However, as remarked by Hunston (2002, p. 26), ''a small corpus can be valuable under certain circumstances'' and one possible circumstance is to construct a specialized corpus for a particular research purpose. There are significant disciplinary variations in the lengths of the introductions, despite their similar numbers of word types (Table 1). Specifically, the average length of article introductions in CL is around 1, 363 words per text, much longer than that of introductions in CE (approximately 900 words per text).

2.2. Key PoS-gram extraction and analysis

As Breeze (2019, p. 81) points out, PoS tag information on a particular corpus is in itself ''meaningless'', and therefore ''a reference corpus needs to be selected in order to conduct a keyness analysis''. While her study focuses on individual

[11] *PLOS ONE* is a peer-reviewed open access journal published by the Public Library of Science since 2006, according to Wikipedia (https://en.wikipedia.org/wiki/PLOS_One). It publishes primary research from a wide range of disciplines within science, engineering, social science and medicine.

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Table 2
Numerical data on the key PoS-grams identified in CCL and CCE.

Corpora	Key PoS-grams	Tokens
CCL	1,368	31,722
CCE	820	17,091

key PoS based on a keyness comparison, the present work extends to investigate key PoS sequences from the two study corpora. To make a cross-disciplinary comparison of key PoS-grams possible, the same reference corpus was selected for them.

While Brett and Pinna (2015) and Pinna and Brett (2018) used the entire 100-million-word BNC corpus as the reference corpus for key PoS-gram analyses, the present study used its written academic component processed by TreeTagger pipeline v2.1 as the reference corpus (see https://app.sketchengine.eu/#dashboard?corpname=preloaded%2Fbnc2_tt21, last retrieved on 25th July, 2020). It is around 17, 627, 082 word tokens (15.69% of the whole BNC).

In the identification and concordance search of key PoS-grams, Sketch Engine with their modified English TreeTagger PoS tagset was adopted (Kilgarriff et al., 2014). This tagset contains 55 tags (see https://www.sketchengine.eu/english-treetagger-pipeline-2/#toggle-id-1, last retrieved on 25th July, 2020), most of which have identified features conventionally considered as parts of speech (e.g., *adverb, adjective* and *modal verb*). A few others such as SYM (symbols, e.g., * *or* =), SENT (sentence break punctuation, e.g.,. *or ?*), FW(foreign word, e.g., *d'hoevre*), CD (cardinal number, e.g., *1, third*), LS (list marker, .e.g., *1), B) or b)*) and UH (interjection, e.g., *uh, oh*) may interfere with the identification of meaningful PoS-grams by SketchEngine (Breeze, 2019, p. 82). Consequently, they were excluded, a practice Breeze (2019) also adopted. However, in her PoS keyness analysis, Breeze additionally discarded the item Z (representing *miscellaneous symbols* in her word), which was not followed here. The reason is that Z stands for possessive ending (e.g., *'s*) rather than miscellaneous symbols in the latest version of modified English TreeTagger PoS tagset (pipeline version 2) developed by SketchEngine, and thus may become an important constituent of PoS sequences.

After rounds of trial analyses and close observation of the data, we decided to set the length of key PoS-grams as six, a number recommended by previous researchers (Brett and Pinna, 2015; Pinna and Brett, 2018; Thompson and Sealey, 2007). A length of five or even fewer may yield an excessive number of PoS-grams, whereas a number of seven or more PoS tags in the sequence might limit the variety of PoS-grams produced and hence the restricted insights (Thompson and Sealey, 2007). Additionally, the maximum length of PoS-grams Sketch Engine could extract is six.

The minimum frequency set for key PoS-gram extraction was 30 per million words and the ''Distribution of Hits'' function offered by Sketch Engine was used to check how key PoS-gram candidates are distributed within the corpus. Those distributed in less than five texts were excluded, to ensure that the key PoS-grams extracted would not be tied to the topic of individual texts. As such, in CCL, two key PoS-grams with their keyness scores originally ranking as the first and the eighth (viz., *IN VVN JJ NN IN NP*[12] and *NN IN VVN JJ NN IN*[13]) were excluded, as all their tokens were found to be in only two and one texts respectively. They all contain the terminology ''*reduced emotional resonance*'' (see their corresponding parts of speech underlined above), which is exactly the topic of the two papers.

The keyness value offered by Sketch Engine is calculated based on a ''simple maths'' procedure (check https://www.sketchengine.eu/documentation/simple-maths/ for the formula and other details), which has resulted in large quantities of results (see Table 2). As such, for the subsequent detailed concordance analysis to derive potential lexicogrammatical frames and phraseological skeletons, only the top ten key PoS-grams were selected for each corpus. Their discursive roles and structure were carefully examined, with special reference to their co-texts, context and the communicative functions of prototypical moves and steps of this conventionalized part-genre, as suggested in the well-known CARS model and its revised version (Swales, 1990, 2004). Finally, the key PoS-grams and their tokens in terms of distribution, structure, meaning and functions were compared across CCL and CCE to discern related cross-disciplinary commonalities and variation.

3. Results and discussion

This part is sub-divided into three sub-sections. In the first two sub-sections, we report on findings from key PoS-gram analyses of CCL and CCE. Following that, a summary of related cross-disciplinary similarities and differences is presented in Section 3.3.

[12] Preposition+Past participle of lexical verb+Adjective+Singular noun+Preposition+Singular proper noun.
[13] Singular noun+Preposition+Past participle of lexical verb+Adjective+Singular noun+Preposition.

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Table 3
Top ten key 6-PoS-grams in CCL.

Key PoS-gram	F1	RF1	F2	RF2	Score	Example token
IN DT JJ NN VBD TO	27	82.5	75	4.3	15.9	*of the present study was to*
NNS VHP VVN IN/that JJ NNS	10	30.6	23	1.3	13.7	*Studies have shown that dyslexic readers*
DT NNS VVP IN/that JJ NN	10	30.6	30	1.7	11.7	*these findings suggest that semantic priming*
VVN TO VV JJ NN NN	13	39.7	45	2.6	11.5	*asked to provide explicit word recognition*
JJ NNS VHP VVN JJ NN	10	30.6	36	2	10.4	*Other studies have provided similar evidence*
JJ NNS VHP VVN IN/that JJ	11	33.6	42	2.4	10.2	*Previous studies have shown that short-term*
DT JJ NN VBD TO VV	27	82.5	151	8.6	8.7	*the present study was to examine*
NN NN VHZ VBN VVN TO	11	33.6	54	3.1	8.5	*adjustment model has been used to*
NN NN NN IN JJ NNS	18	55	108	6.1	7.9	*word segmentation processing of Chinese readers*
DT JJ NN JJ NN NN	11	33.6	62	3.5	7.7	*the masked priming lexical decision task*

Notes: 1. F1 and F2 indicate raw frequencies of each key PoS-gram in CCL and the reference corpus respectively.
2. RF1 and RF2 represent their relative frequencies in CCL and the reference corpus respectively (per million words).

3.1. Salient PoS-grams in CCL

Table 3 shows the top ten key PoS-grams in CL introductions, with their raw frequencies, relative frequencies in the two corpora, key scores and example tokens listed. The PoS-gram with the highest key score is *IN DT JJ NN VBD TO*[14] (82.5 times per million words), which together with *DT JJ NN VBD TO VV*,[15] ranking as the seventh, are found to be sub-grams of the 7-PoS-gram (*IN DT JJ NN VBD TO VV*[16]). The numbers of their tokens are equal (27) (see Table 3), among which 25 concordance lines are overlapping. Nevertheless, all of their tokens function similarly to announce research purposes or aims and a lexicogrammatical frame can thus be derived, i.e., ''{The/A/One} (primary/main/first/second) {aim/goal/purpose/focus} of the {present/current} {study/experiment/investigation} was *to*-infinitive clause'' (Appendices A and B).

The words enclosed within curly brackets represent a range of lexical choices in the slot which often belong to the same semantic set, resulting in ''the formation of a series of loosely synonymous expressions'' (Philip, 2008, p. 99). The curly brackets also indicate word choice within them being obligatory, i.e., at least one word is supposed to be selected, whereas the words shown in parentheses are all optional. This difference is intended to capture the different importance of components within patterns, giving readers a clear sense of overall patterning and internal variability. All the tokens of these two key PoS-grams are closely connected to ''outlining purposes/announcing present research'', the only obligatory step under Move 3 ''occupy the niche'' in Swales's (1990, 2004) CARS model. This finding consolidates the importance of announcing present research descriptively or purposively in the opening phrase of the article (Bondi, 2010) whilst evidencing a high degree of formulaicity in language realizations of this step (Cortes, 2013). Such arrays of exponents of the two key PoS-grams mapped with this communicative function would be directly helpful to novice writers.

Another set of four key PoS-grams (viz., *NNS VHP VVN IN/that JJ NNS*,[17] *JJ NNS VHP VVN IN/that JJ*,[18] *DT NNS VVP IN/that JJ NN*,[19] and *JJ NNS VHP VVN JJ NN*[20]) with partly analogous structure are found to play similar textual functions—summarizing previous studies or synthesizing findings in the literature (Table 3). A close examination of their concordance lines reveals that they mostly relate to the essential step of ''reference to previous research or scholarship (normally more than one author) to make topic summarization'' suggested in Swales's (1990) CARS model, as illustrated below:

(1) **Studies have shown that dyslexic readers** elicited smaller N170 amplitudes compared to regular readers. . . [35]–[36]. (*NNS VHP VVN IN/that JJ NNS*)

(2) . . .**previous studies have shown that positive** or negative words, or both, elicited a greater late positive complex. . . [3]–[9]. *(JJ NNS VHP VVN IN/that JJ)*

[14] Preposition+Determiner+Adjective+Singular noun+Verb *BE* in the past tense+*TO* infinitive.
[15] Determiner+Adjective+Singular noun+Verb *BE* in the past tense+*TO* infinitive+Base form of lexical verb.
[16] Preposition+Determiner+Adjective+Singular noun+Verb *BE* in the past tense+*TO* infinitive+ Base form of lexical verb.
[17] Plural noun+Non-3rd person singular present form of the verb *HAVE*+Past participle of lexical verb+*That* as subordinator+Adjective+Plural noun.
[18] Adjective+Plural noun+Non-3rd person singular present form of the verb *HAVE*+Past participle of lexical verb+*That* as subordinator +Adjective.
[19] Determiner+Plural noun+Non-3rd person singular present form of lexical verb+*That* as subordinator+Adjective+Singular noun.
[20] Adjective+Plural noun+Non-3rd person singular present form of the verb *HAVE*+Past participle of lexical verb+Adjective+Singular noun.

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(3) **These data suggest that whole word** representations exist and can affect recognition. . . [23], [24]. (*DT NNS VVP IN/that JJ NN*)

(4) **Previous studies have used demographic information**. . .to explain the variability in CI outcomes, but with limited success [7], [8]. (*JJ NNS VHP VVN JJ NN*)

In particular, the tokens of the former two PoS-grams yield the lexicogrammatical frame ''{studies/works} have {shown/mentioned/demonstrated/found/reported} *that*-clause'', with the verb *show* most often used (5 out of the 10 and 11 times respectively). These two PoS-grams only differ in the modifier preceding the first NNS and the composition of the noun phrase as the subject of *that*-clause. Specifically, for *NNS VHP VVN IN/that JJ NNS*, not only adjectives such as *previous, several* and *other*, but determiners like *these* and *some* could be used preceding the first NNS (tokens being either *studies* or *works*) to summarize the studies reviewed or to synthesize sources for a contrast (e.g., *other studies have shown that*. . .). As for *JJ NNS VHP VVN IN/that JJ*, the structure after the subordinator *that* is not complete and *JJ* is only one of the multiple modifiers of the head noun as the subject, e.g., *positive or negative words (**JJ** CC JJ NNS)* and *individual Chinese characters (**JJ** JJ NNS)*.

Functioning similarly to summarize the literature, the key 6-PoS-gram *DT NNS VVP IN/that JJ NN* suggests another lexicogrammatical frame ''{the/these} {findings/results/data} {suggest/indicate} *that*-clause'' (Appendix C). It is noted that only a very restricted set of verbs (i.e., *suggest*, *indicate*) in the simple present tense can be used in this pattern, with the subject being *findings, results* or *data*, indicating the validity of generalisations of previous findings. This is in stark contrast with the lexicogrammatical frame ''{studies/works} have {shown/mentioned/demonstrated/found/reported} *that*-clause'' we just discussed, where another set of verbs suggesting research or discursive acts (e.g., *show, demonstrate*, *find, report*) are used in the present perfect tense, which tend to be co-selected with the subject *studies/works* as the agent. These two contrasting co-selection patterns with analogous functions but fine-grained internal structural and compositional variations might not have been fully detected, if we only study salient individual words or PoS tags or multi-word sequences in isolation. This can clearly demonstrate the unique strengths of the PoS-gram analysis in grouping patterns and showing variability within patterns.

Within this set of four salient PoS-grams for topic summarization, the last variant *JJ NNS VHP VVN JJ NN* involves using the present perfect ''to make a general statement about the state of research activity in a given area'' (Collins Cobuild English Grammar, 4th edition, 2017, p. 1102), which resembles the aforementioned PoS-grams *NNS VHP VVN IN/that JJ NNS* and *JJ NNS VHP VVN IN/that JJ*.

Yet, its difference lies in not having a *that*-clause but a noun phrase in post-predicate position, which could be classified into two main categories, as its concordance lines suggest: (1) a group of noun phrases with more abstract head nouns denoting results, effects or a resulting phenomenon (70%) (Examples 5–6), and (2) others referring to a specific research variable or target, which often collocate with verbs denoting research procedures or acts (30%) (Example 4). The former type obviously concerns what has been achieved, whereas the latter category relates to what has been done.

(5) **Other studies have provided similar evidence,** . . . (see [3], [11] for reviews). (*JJ NNS VHP VVN JJ NN*)

(6) Indeed, **many studies have demonstrated N400 mismatch** effects. . . (for example [21,22]). (*JJ NNS VHP VVN JJ NN*)

Further, regarding another salient 6-PoS-gram *VVN TO VV JJ NN NN*,[21] four out of its 13 tokens structured in the form of the past simple passive plus the *to*-infinitive have been used towards the end of introductions to realize the step of ''summarizing the methods used'', as indicated in Swales's (2004) revised CARS model. The occurrence of this step is more likely in papers ''whose principal outcome can be deemed to reside in their methodological innovations'' (Swales, 2004, p. 231). One such example is provided below.

(7) . . .participants in the present study were also **asked to provide explicit word recognition** judgements at the end of each trial, making it possible to. . .. (*VVN TO VV JJ NN NN*)

Another five tokens of this PoS-gram, if we expand their left context, are noted to be in the structure of the present perfect passive plus the *to*-infinitive. They all function to state synthesized findings, often accompanied with multiple references encapsulated in a non-integral citation plus reporting verbs such as *show, demonstrate* and *suggest*, as illustrated in Example (8):

[21] Past participle of lexical verb+*TO* infinitive+Base form of lexical verb+Adjective+Singular noun+Singular noun.

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(8) The incorporation of both serial and lexicalized varieties of fingerspelling in elementary reading instruction has been **demonstrated to enhance English vocabulary acquisition** in deaf children [35,36, 37]. (*VVN TO VV JJ NN NN*)

Functioning differently, the remaining four tokens are parts of the structure— ''the lexical verb in the simple passive plus the *to*-infinitive'', with two using verbs in the present tense to indicate common research practices or generally accepted knowledge (Example 9). The other two use verbs in the past simple passive to review individual prior studies with probably inconclusive findings (Example 10).

(9) As noted by some researchers [21], this type of test is routinely **used to assess oral language comprehension** in children with specific language impairments. (*VVN TO VV JJ NN NN*)

(10) However, early vocabulary level (assessed at the beginning of kindergarten) was **found to predict early reading comprehension** performances . . . in other studies with English children [13]. (*VVN TO VV JJ NN NN*)

The key PoS-gram *NN NN VHZ VBN VVN TO*[22] contains a ''NN NN'' construction as its subject, e.g., *adjustment model* (Table 3) and *negativity bias* (Example 11). Functionally, it might arguably be considered as another variant of the set of PoS-grams for synthesizing findings (e.g., *NNS VHP VVN IN/that JJ NNS*). However, its difference resides in the subject being a concrete research target or item (e.g., *negativity bias* in Example 11) rather than a general term like *studies/findings/results/data* used in tokens of the four 6-PoS-grams for topic summarization. This perhaps necessitates the use of the passive plus the *to*-infinitive to indicate summarized results.

(11) In middle-aged and older adults, the **negativity bias has been found to** be reduced [13–14] . . .. (*NN NN VHZ VBN VVN TO*)

Concerning the last two key PoS-grams in Table 3 (*NN NN NN IN JJ NNS*[23] and *DT JJ NN JJ NN NN*[24]), their grammatical structure may embody the preferential use of nominalization in academic register (Biber et al., 1999). Notice that the former PoS-gram *NN NN NN IN JJ NNS* comprise a prepositional phrase (*IN JJ NNS*) postmodifying the noun phrase (*NN NN NN*) to give extra or specific information about the head noun (mainly possession and identifying features). The head noun in these tokens is invariably a discipline-specific specialized term such as *speech act processing, discovery rate correction* and *word segmentation processing* (Table 3). The tokens of the latter PoS-gram *DT JJ NN JJ NN NN* consistently contain the recurrent terminology ''priming lexical decision task/experiment'' (72.7%) and accordingly a simple lexicogrammatical frame could be derived, i.e., ''{a/the/an} {masked/affective} priming lexical decision {task/experiment/paradigm}'', based on 9 out of the 11 tokens. As such, both PoS-grams feature CL introductions with their tokens frequently suggestive of disciplinary content.

3.2. Salient PoS-grams in CCE

Table 4 lists the top ten salient PoS-grams in CCE, their key value and example tokens, as compared to the written academic component of BNC. In contrast to CCL, the 6-PoS-gram with the highest key score identified in CCE is *DT NN VBZ VVN RB VVZ*[25] (see concordance lines in Appendix D), whose tokens are structure-outlining sentences, ''nearly always a final element in Move 3'' of the CARS model (Swales, 2004, p. 232). While the use of this element seems absent in the disciplines with an established IMRD-like sectional arrangement such as biochemistry studied in Kanoksilapatham (2003), ''in other fields that lack such an arrangement, such as computer science, information science, biostatistics, or economics, this structure-outlining option becomes close to obligatory'' (Swales, 2004, p. 232). Given that CE is close to the field of computer science, Swales's observation has well accounted for the strong presence of this PoS-gram for roadmapping the structure of the paper in CCE. The head verb ''*organize*'' or ''*structure*'' has been constantly used in the present passive form and a lexicogrammatical frame could be derived: The {reminder/rest} (of) (the/this) (paper/article) is {organized/structured} as follows.

Indeed, this lexicogrammatical frame can be represented in a semantic sequence, i.e., ''outline introductory formulae'' proposed by Bondi (2010): DISCOURSE UNIT (e.g., this paper/the rest of the paper) + V-STRUCTURE (e.g., is organized/is structured) + CATAPHORA (e.g., as follows/in the following way).

[22] Singular noun+Singular noun+3rd person singular present form of the verb *HAVE*+Past participle of the verb *BE*+Past participle of lexical verb +*TO* infinitive.
[23] Singular noun+Singular noun+Singular noun+Preposition+Adjective+Plural noun.
[24] Determiner+Adjective+Singular noun+Adjective+ Singular noun+Singular noun.
[25] Determiner+Singular noun+3rd person singular present form of the verb *BE*+Past participle of lexical verb+Adverb+3rd person singular present form of lexical verb.

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Table 4
Top ten key 6-PoS-grams in CCE.

Key PoS-gram	F1	RF1	F2	RF2	Score	Example token
DT NN VBZ VVN RB VVZ	15	69.9	5	0.3	55.3	*This paper is organized as follows*
NN NNS VHP VBN VVN TO	10	46.6	38	2.2	15.1	*computer programs have been developed to*
NP NP NP NP NP NP	76	354.3	555	31.5	10.9	*Python Version Numpy Version Scipy Version*
JJ JJ NN NN VVN IN	10	46.6	60	3.4	10.8	*stochastic local search algorithm based on*
IN DT JJ NN VBD TO	11	51.3	75	4.3	9.9	*of the present study was to*
JJ NN NNS VHP VBN VVN	16	74.6	128	7.3	9.2	*several software applications have been proposed*
IN JJ NN NNS JJ IN	11	51.3	89	5	8.6	*of common simulation languages such as*
IN JJ CC JJ NN NN	13	60.6	116	6.6	8.1	*with free and open source software*
TO VV JJ NN NNS IN	15	69.9	143	8.1	7.8	*to provide adequate download speeds for*
DT JJ NN VBD TO VV	15	69.9	151	8.6	7.4	*the present study was to investigate*

Notes: 1. F1 and F2 indicate raw frequencies of each key PoS-gram in CCE and the reference corpus respectively.
2. RF1 and RF2 represent their relative frequencies in CCE and the reference corpus respectively (per million words).

Further, a close look at all concordance lines of this pair of key PoS-grams (*NN NNS VHP VBN VVN TO*,[26] ranking the second, and *JJ NN NNS VHP VBN VVN*,[27] ranking the sixth) has revealed that five of their concordance lines are identical, from which a longer 7-PoS-gram could be derived, viz., *JJ NN NNS VHP VBN VVN TO*,[28] as illustrated in Example 12.

(12) **Several computer programs have been developed to** analyze relaxation data and generate the parameters. . ., including Modelfree [4], relax [5] and MOLDYN [6]. (*JJ NN NNS VHP VBN VVN TO*)

The rest of their concordance lines has revealed slightly different co-occurrence patterns for the two PoS-grams. For instance, different from the PoS-gram *NN NNS VHP VBN VVN TO* which suggests only the *to*-infinitive clause used immediately after the present perfect passive, the prepositional phrase is equally often used (as the *to*-infinitive clause) after the present perfect passive for *JJ NN NNS VHP VBN VVN* (31.3%), as exemplified in the underlined part of Example (13):

(13) Previous studies [7,8] have indicated that pattern matching consumes approximately 70% of system execution time. Many software- and **hardware-centered pattern-matching algorithms have been proposed** for NIDSs. (*JJ NN NNS VHP VBN VVN*)

Despite this slight difference in collocational patterns, all tokens of both PoS-grams with the shared use of present perfect passives have uniformly functioned to synthesize the current state of knowledge or advances on computer programs, software applications or computational methods (see Examples 12–13). Consequently, a lexicogrammatical frame can be extracted for most concordance lines of these two PoS-grams: [computer programs/algorithms/models] have been {developed/proposed/employed/used/applied/shown} (*to*-infinitive clause or prepositional phrase). Note that the subject within the square brackets represents a recurring semantic element rather than concrete lexical choices, similar to ''*meaning elements*'' in Hunston's (2008) definition of ''*semantic sequences*''. This phraseological pattern for the function of topic summarization embodies discipline-specific features, in that the subject has been specifically relevant to computer programs/algorithms/models and the collocational verbs following are dominantly *develop* or *propose* (together over 50% for both PoS-grams).

Another noticeable disciplinary feature is apparently more compact language use than in CCL, as evidenced in the intensive use of pre-modifications and post-modifications of noun phrases, and the total absence of the *that-clause* but strong presence of the *to-infinitive clause* in top-ranking key PoS-grams. To illustrate, the presence of the construction ''noun + noun(+noun) . . .'' has been found in seven out of the 10 top-ranking key PoS-grams (Table 4). Additionally, a combination of multiple adjectives and nouns could also serve as premodifiers, as seen from the concordance lines of another PoS-gram *JJ JJ NN NN VVN IN*[29] in Appendix E.

[26] Singular noun+Plural noun+Non-3rd singular present form of the verb *HAVE*+Past participle of the verb *BE*+Past participle of lexical verb+*TO* infinitive.
[27] Adjective+Singular noun+Plural noun+Non-3rd singular present form of the verb *HAVE*+Past participle of the verb *BE*+Past participle of lexical verb.
[28] Adjective+Singular noun+Plural noun+Non-3rd singular present form of the verb *HAVE*+Past participle of the verb *BE*+Past participle of lexical verb+*TO* infinitive.
[29] Adjective+Adjective+Singular noun+Singular noun+Past participle of lexical verb+Preposition.

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Further, the key PoS-gram ''*NP NP NP NP NP NP*'' ranking the 3rd in the list is noted to have as many as 76 occurrences (Table 4). While it is an important characteristic of academic writing to use one or multiple nouns to premodify a head noun (Biber et al., 1999), ''using more than four premodifiers makes the noun phrase difficult to understand, especially when this consists only of nouns'' (Collins Cobuild English Grammar, 4th edition, 2017, p. 1098). Nevertheless, all tokens of the key PoS-gram ''*NP NP NP NP NP NP*'' represent discipline-specific technical terms, with most being the names of operating systems Python version plus a couple the names of regulations (Example 14), which may be another marked disciplinary feature in language use.

(14) . . .the recently enacted **EU General Data Protection Regulation EU-2016**/679 (GDPR) not only raised . . .. (*NP NP NP NP NP NP*)

In addition to pre-modifications, the PoS-gram *JJ JJ NN NN VVN IN* also suggests the -ed participle clause as a common way of post-modifying the noun phrase, as exemplified in 90% of its concordance lines in Appendix E. Moreover, the tokens of another three PoS-grams *TO VV JJ NN NNS IN*[30] (80%), *IN JJ NN NNS JJ IN*[31] (near 60%) and *IN JJ CC JJ NN NN*[32] (near 50%) have been dominantly post-modifiers of a noun phrase, as illustrated respectively in Examples 15–17. Especially, the former two are suggestive of incomplete structures on both ends. According to *Collins Cobuild English Grammar* (the 4th edition, 2017), participle and infinitive clauses and prepositional phrases are often used to reduce the relative clause in academic writing to make language compressed whilst giving more information about the noun phrase they modify. As for the rest few tokens of *TO VV JJ NN NNS IN*, they are parts of the *to*-infinitive structures entailed by verbs such as *attempt* (2 times) and *continue* (1 time). In contrast, the rest of the tokens for the PoS-gram *IN JJ NN NNS JJ IN* serve as post-modifiers of a verb phrase in passive voice. This also applies to the PoS-gram *IN JJ CC JJ NN NN* for most of its other tokens not post-modifying a noun phrase (Example 18).

(15) . . . a novel research direction is to investigate uncompensated samples as a way **to conduct large scale studies with** the benefit of being cheaper and better representative populations [1,2]. (*TO VV JJ NN NNS IN*)

(16) The Pipeline's task-manager provides. . ., and integrates the direct and batch processing capabilities **of available grid-management environments such as** Oracle Grid Engine. . .. (*IN JJ NN NNS JJ IN*)

(17) Concepts **of model-based and model-free reinforcement learning** are incorporated. . .. (*IN JJ CC JJ NN NN*)

(18) . . .several such orphan crops have been enriched **with full or partial reference genome** sequence information. (*IN JJ CC JJ NN NN*)

Despite remarkable discipline-specific features, the concordance lines of two salient PoS-grams (''*IN DT JJ NN VBD TO*''[33] and ''*DT JJ NN VBD TO VV*''[34]) have been consistently found to be purposive statements in CCE as well (see Appendices F and G). A lexicogrammatical frame could thus be yielded, i.e., ''{The/A} (main/major/first/second/secondary) {objective/goal/aim/purpose} (of/for) (the/this) (present/current) (study/review) was *to*-infinitive clause''. Among their concordance lines, quite a majority are overlapping, which has implied the existence of a longer 7-PoS-gram (''*IN DT JJ NN VBD TO VV*''[35]). As suggested earlier, this is concomitant with the CARS model that indicates ''announcing research descriptively/purposively'' to be an obligatory step in introductions (Swales, 1990, 2004). The relatively high degree of formulaicity in language realization of this step has again been confirmed in CE introductions.

3.3. Cross-disciplinary similarities and differences in salient PoS-gram sequences

The preceding two sub-sections have presented detailed findings on key PoS-grams in CCL and CCE, with scattered mentioning of related cross-disciplinary commonalities and differences. As introductions to RAs have been acknowledged as a largely conventionalized part-genre (Lin, 2020; Swales, 1990, 2004), what is shared across the two corpora in their PoS-gram analyses might be considered as features of language patterning specific to this part-genre and what distinguishes one from the other might be characterizing the particular discipline. The major cross-disciplinary commonalities and differences are summarized below. First of all, the PoS-grams with high keyness scores have been successfully identified for introductions of both disciplines, with their representative lexicogrammatical frames and phraseologies highlighted, which has empirically validated the phraseological tendency and idiomaticity of language use

[30] *TO* infinitive+Base form of lexical verb+Adjective+Singular noun+Plural noun+Preposition.
[31] Preposition+Adjective+Singular noun+Plural noun+Adjective+Preposition.
[32] Preposition+Adjective+Coordinating conjunction+Adjective+Singular noun+Singular noun.
[33] Preposition+Determiner+Adjective+Singular noun+The past tense form of the verb *BE*+*TO* infinitive.
[34] Determiner+Adjective+Singular noun+The past tense form of the verb *BE*+*TO* infinitive+Base form of lexical verb.
[35] Preposition+Determiner+Adjective+Singular noun+The past tense form of the verb *BE*+*TO* infinitive+Base form of lexical verb.

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Table 5
Lexicogrammatical frames for purposive-announcement.

Corpus	Pattern	Examples	Frequency
CL	{The/A/One} (primary/main/first/second) {aim/goal/purpose/focus} of the {present/current} {study/experiment/investigation/review} was *to*-infinitive clause	**- *The primary aim of the present study*** *was to test*... - ***The purpose of the present study*** *was to explore*...	29
CE	{The/A} (main/major/first/second/secondary) {objective/goal/aim/purpose} (of/for) (the/this) (present/current) (study/review) was *to*-infinitive clause	**-*The goal of the current study*** *was to extend*... **-*The secondary objective*** *was to test*...	16

in academic genres (Sinclair, 1996; Gledhill, 2000). Second, in both CCL and CCE, the key PoS-grams with their discursive functions matching with the steps of ''purposive announcement'' and ''topic summarization/synopsis of previous findings in the literature'' suggested in the CARS model (Swales, 1990, 2004) have been identified. This has not only evidenced the importance of these two functional steps in the introductions, but has foregrounded a high level of formulaicity in language use to realize them.

Regarding salient PoS-grams for realizing them, the couple associated with ''purposive announcement'' (viz., *IN DT JJ NN VBD TO* and *DT JJ NN VBD TO VV*) is shared in both corpora. In line with this, the lexicogrammatical frames identified for purposive announcement in the introductions of the two disciplines resemble each other very closely (see Table 5).

Yet, the key PoS-grams identified for realizing the step of ''topic summarization/synopsis of previous findings in the literature'' are not identical for the two disciplines. As pointed out before, in CCL, the subjects tend to be words such as ''*studies, works*'' (co-selecting with verbs like ''*show, demonstrate*'' in the present perfect tense and active voice) or ''*findings, results and data*'' (collocating with verbs ''*suggest, indicate*'' in the simple present tense and active voice). By contrast, in CCE, the subjects are words related to computer programs or software applications, often being particular algorithms or modeling frameworks, and their collocational verbs are dominantly *develop* or *propose* used in the present perfect tense and passive voice, which manifests its distinctive disciplinary nature. In all, two lexico-grammatical frames could be synthesized for CL introductions in contrast to one for CE introductions for the function of ''topic summarization or synopsis of previous findings'' (see Table 6). Such different language patterns for realizing the same functional step across two contrasting disciplines should be of particular attention to novice writers.

Another difference worth noting is the total absence of the *that*-clause in all salient PoS-grams identified for CE introductions, indicating compressed language use in this discipline, which contrasts with the more substantial presence of the *that*-clause in CCL (e.g., present in three out of the four key PoS-grams for realizing the topic summarization step). As detailed earlier, the intensive use of the construction ''noun + noun(+noun) . . .'' as well as pre-modifications and/or post-modifications of noun phrases, and the frequent adoption of the *to*-infinitive clause to replace the *that*-clause have all contributed to obviously more compact language use in CE introductions. In addition, only CE introductions are distinctively featured with the strong presence of the PoS-gram ''*DT NN VBZ VVN RB VVZ*'' for outlining paper structure, which however is not a trend in CL. Finally, discipline-specific features are also manifested in certain key PoS-grams identified, such as the salient PoS-gram ''*NP NP NP NP NP NP*'' with 76 occurrences in CCE, whose tokens are all the names of computer operating systems and regulations.

Table 6
Lexicogrammatical frames for topic summarization or synopsis of previous findings.

Corpus	Pattern	Examples	Frequency
CL	{studies/works} have {shown/mentioned/demonstrated/found/reported} *that*-clause or noun phrase	***-previous studies have demonstrated that*** *emotional*... ***-Other studies have provided similar evidence***...	26
	{the/these} {findings/results/data} {suggest/indicate} *that*-clause	***-These results suggest that*** *lexical processing*... ***-these findings indicate that*** *ordinal processing*...	10
CE	[computer programs/algorithms/models] have been {developed/proposed/employed/used/applied/shown} (*to*-infinitive clause or prepositional phrase)	-. . .*various* ***dual-system frameworks have been proposed*** *[5,7].* -*several* ***software applications have been proposed to*** *independently evaluate selection pressures at the codon-level [15]–[18]*	21

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4. Conclusion

This paper is unique in setting PoS-grams as the unit of analysis in a most conventionalized research part-genre across disciplines. It is, to the best of our knowledge, the first cross-disciplinary study of key PoS-grams in RA introductions. This study has clearly demonstrated the huge potential of the PoS-gram procedure in unfolding language patterning and variability in specialized (part-)genres (Brett and Pinna, 2015). It opens up a new way for examining discipline-specific and/or genre-specific (patterned) language use and discursive features and the related variations and commonalities thereof for EAP researchers and practitioners.

To sum up, the present study has identified salient PoS-grams in research introductions from two contrasting disciplines (viz., CL and CE), which could be generally classified into three categories. The first category includes key PoS-grams with shared functions and identical composition and patterns in introductions from both disciplines, e.g., *IN DT JJ NN VBD TO* and *DT JJ NN VBD TO VV,* associated with the functional step ''purposive announcement''. The second category contains those with overlapping/similar discursive functions but varying composition and patterns in CCL and CCE. For example, a set of four key PoS-grams (viz., *NNS VHP VVN IN/that JJ NNS*, *JJ NNS VHP VVN IN/that JJ*, *DT NNS VVP IN/that JJ NN*, and *JJ NNS VHP VVN JJ NN*) identified in CCL together with another couple of key PoS-grams (viz., *NN NNS VHP VBN VVN TO* and *JJ NN NNS VHP VBN VVN*) in CCE play similar textual functions, i.e., to summarize previous studies or synthesize findings in the literature. However, a glimpse of their associated lexicogrammatical frames would reveal interesting but important differences in their subjects, the voice and aspect of the main verbs and their co-selected structure (i.e., whether to use the *that*-clause, the *to*-infinitive clause or the prepositional phrase). To detail, in CCL, the subjects tend to be either words like ''*studies, works*'' (co-selecting with verbs such as ''*show, demonstrate*'' in the present perfect tense and active voice) or ''*findings, results and data*'' (co-occurring with verbs like ''*suggest, indicate*'' in the simple present tense and active voice). By contrast, in CCE, another regular pattern is generated for topic generalization/synopsis of previous findings. The subjects are found to be terms relative to computer programs, model frameworks, algorithms or software applications and their collocational verbs are dominantly *develop* or *propose* used in the present perfect tense and passive voice. Therefore, even for realizing the same functional step in a largely conventionalized part-genre (e.g., the step of ''topic summarization/synopsis of findings in the literature'' as discussed here), writers from different disciplinary discourse communities have their own distinctive patterns to use.

The final category comprises the other key PoS-grams uniquely found in either CCL or CCE. Within this group, some may be directly reflective of disciplinary nature, e.g., the PoS-gram with the highest key value identified in CCE is associated with ''*outlining the structure of articles*'', an optional element only favored in introductions from a few disciplines without an established IMRD-like sectional arrangement such as CE (Kanoksilapatham, 2003; Swales, 2004).

In addition to identifying sets of characteristic lexicogrammatical frames and phraseologies that could be directly transformed into EAP pedagogical input, the PoS-gram analysis has also helped revealing contrasting language styles in introductions of the two disciplines. The apparently more compact language use has been noted in CE introductions than in CL introductions, as evidenced in the total absence of the *that*-clause but the strong presence of the *to*-infinitive clause and the prepositional phrase instead in tokens of top-ranking key PoS-grams identified in CCE. Contrastingly, in CCL, the use of the *that*-clause is far more frequent, e.g., three out of the four key PoS-grams for realizing the step of topic summarization do contain it. The more compressed language style of academic introductions in CE could also be perceived from the particularly intensive use of the construction ''noun + noun(+noun) . . .'' as well as the pre-modifications and/or post-modifications of noun phrases in them.

All these findings have important implications for the teaching and learning of academic writing in English. The results of this study could be utilized for the pedagogy of teaching EAP for computer engineering and linguistics students in terms of the selection of typical grammatical patterns, lexicogrammatical frames and phraseology for teaching and the demonstration of language patterning and variability via the PoS-gram procedure. In the light of this, language focus in teaching research writing is supposedly not just on move-specific individual linguistic signals or particular language features, but on salient language co-selection patterns to facilitate fluency, naturalness and effectiveness in student academic language use. Meanwhile, teaching activities such as asking students to fill in the missing words of a given pattern need to be devised to highlight lexical variants in certain PoS position, so that students could pay due attention to fixedness and variability within the patterns.

Further, some salient language patterns represented by lexicogrammatical frames and their associated phraseologies, as revealed by the present PoS-gram analysis, are closely linked to particular communicative functions of the (part-) genre. Nonetheless, even when realizing the same functional move or step, different disciplinary discourse communities may have their own preferential co-selected linguistic patterns, a point that should be addressed in teaching to enhance novice writers' understanding of the rhetoric, genre and disciplinary conventions.

The utility of the corpus-based PoS-gram technique combined with genre study awaits more systematic explorations, especially by EAP teachers who promote data-driven learning (Charles, 2014; Otto, 2021). Since the corpus-based PoS-

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gram analysis could be a promising line of inquiry into language patterning in the EAP/ESP world, it needs to be extended to other part-genres of the RA to arrive at a full description of the most typical expressions for this genre, as well as to other specialized genres for academic and professional communication.

Conflicts of interest

None declared.

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Appendix A. Sample concordance lines of *IN DT JJ NN VBD TO* in CCL

Left context	KWIC	Right context
lopment or nutritional status [19,24]. The aim	**of the current study was to**	test whether the stimulation intervention had
with other components of language. The aim	**of the present experiment was to**	examine language lateralization in a group of
complex figure [41]. Therefore, the main aim	**of the current study was to**	better characterize early activation of IFG - s
ugh the paper or display ([17], [18]). The aim	**of the present study was to**	examine the flexibility of the input coding sch
ts reminding patients of task goals. The aim	**of the present study was to**	investigate whether priming could be used to
al to respond quickly and accurately, the aim	**of the present experiment was to**	investigate whether priming can be used to e
d to be tested in modern times. The purpose	**of the present study was to**	explore whether the segmentation rules of C
ven research question [30]. The primary aim	**of the present review was to**	examine the evidence for individual CBAT as
ts diagnosed with ADHD. Therefore, the aim	**of the present study was to**	assess source discrimination in adult patients
ables of the target words. Therefore, the goal	**of the present study was to**	reevaluate syllabic segmentation in light of th
R using a lowest WN level of 65 dB. The goal	**of the current study was to**	test the hypothesis that different intensities o
ecall of the context word. Thus, the first goal	**of the current investigation was to**	test the viability of the integration-elaboration
he present study. The present study The aim	**of the present study was to**	examine the contributions to reading compre
y been partially processed. The primary aim	**of the present study was to**	test the early speech act recognition account

NoteIn all, there are 27 occurrences of this PoS-gram. For a better visual display, a sample of about half of the concordance lines is presented.

Appendix B. Sample concordance lines of *DT JJ NN VBD TO VV* in CCL

Left context	KWIC	Right context
re needed. The present study The aim of	**the present study was to examine**	both learning and retention in children with
g ‘‘online’’ comprehension. The first aim of	**the current study was to evaluate**	comprehension of words across various s
es discover that they had lied. The goal of	**the present research was to examine**	if speakers indeed distance themselves fro
ther components of language. The aim of	**the present experiment was to examine**	language lateralization in a group of patien
review, see [11; 39–41]). One purpose of	**the present study was to apply**	our knowledge of current research on met
the CERAD-NP battery. Hence, the aim of	**the present study was to investigate**	possible differences in cognitive profile bet
gnosed with ADHD. Therefore, the aim of	**the present study was to assess**	source discrimination in adult patients with
of the target words. Therefore, the goal of	**the present study was to reevaluate**	syllabic segmentation in light of this hypot
he paper or display ([17], [18]). The aim of	**the present study was to examine**	the flexibility of the input coding scheme fo
esearch question [30]. The primary aim of	**the present review was to examine**	the evidence for individual CBAT as an eff
g a lowest WN level of 65 dB. The goal of	**the current study was to test**	the hypothesis that different intensities of
of the context word. Thus, the first goal of	**the current investigation was to test**	the viability of the integration-elaboration a
sent study. The present study The aim of	**the present study was to examine**	the contributions to reading comprehensio

NoteSimilar to Appendix A, around half of all the 27 concordance lines are listed for a better visual display.

Appendix C. Concordance lines of *DT NNS VVP IN/that JJ NN* in CCL

Left context	KWIC	Right context
performance at the task. Together,	the data suggest that declarative memory	deficits in SLI may be due largely, i
with our current hypothesis. Also,	these findings indicate that ordinal processing	has a biological base and, hence,
ng conditions [10]. Taken together,	these findings suggest that linguistic knowledge	significantly contributes to the resto
al preview benefit. Taken together,	these findings suggest that phonological recoding	is an automatic process at some s
ocabularies do not [7,8]. Together,	these findings suggest that semantic priming	is driven by several factors includin
context of a word than in isolation.	These data suggest that whole word	representations exist and can affec
red to less skilled readers [10, 11].	These findings indicate that extra-foveal information	is an important determinant of read
unrelated uncrowded prime word.	These findings indicate that semantic information	is activated even when a word is n
ss-language activation either [33].	These results suggest that language-specific speech	cues were exploited by bilingual list
mental power is sufficient [20, 43].	These results suggest that lexical processing	does not require central attention.

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Appendix D. Concordance lines of *DT NN VBZ VVN RB VVZ* in CCE

Left context	KWIC	Right context
rful for modeling open source	This paper is organized as follows	. In the next section, script-based
labware.		FOSS
er improve the classifier	The paper is organized as follows	. Section Materials and Methods
performance).		provide
is the basis for Section 1 of this	The remainder is organized as follows	: Section 2 reviews related work;
paper.		Sectio
rful for modeling open source	This paper is organized as follows	. In the next section, script-based
labware.		FOSS
ce analysis of the controller. The	the paper is organized as follows	: Section 2 describes work related to
rest of		the
al are also interchangeable. The	this paper is organized as follows	. In section two, we illustrate the
rest of		concep
iated with uncertainty. The	this paper is organized as follows	. The Material and Methods section
remainder of		incl
tive functions are the same. The	the paper is organized as follows	. In the next section, we first give a
rest of		non-
development of IQM, the	this paper is organized as follows	. We review existing open source
remainder of		applic
results are compared. The	this paper is organized as follows	. In the Materials and Methods
remainder of		section,
g such functionalities. The	the paper is structured as follows	: while Section 2 introduces our
remainder of		novel G
ce analysis of the controller. The	the paper is organized as follows	: Section 2 describes work related to
rest of		the
ciences, see e.g., [2, 12–19]The	the paper is organized as follows	. Next section shows the main
rest of		methods
agram of proposed method. The	the paper is organized as follows	, section III summarize the existing
rest of		meth
atching search terms. The	this article is organized as follows	. Section 1 details the
remainder of		methodological b

Appendix E. Concordance lines of *JJ JJ NN NN VVN IN* in CCE

Left context	KWIC	Right context
ow renamed Bi-Manu Trainer), a	game-based virtual reality system designed for	upper-limb rehabilitation, has bee
im of this study was to design an	on-line fNIRS-based inference system dedicated to	: estimating the pilot's state (perfo
x neural activity into a correlated	mechanized prosthetic arm movement used for	self-feeding [4]. Other than these
e Virtual Operative Assistant) for	complex technical task enhancement based on	explainable machine learning, and
e Virtual Operative Assistant) for	complex technical task enhancement based on	explainable machine learning, and
essing units (GPUs), which have	superior parallel processing power compared to	CPUs, are likely candidates for int
ty or takes too long. In sports, the	likely first transfer study focused on	simulator training for playing table
sm. In Section 6, we propose the	stochastic local search algorithm based on	QCCA heuristic. In Section 7, we

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deployment of new services. The	novel centralized management paradigm introduced by	SDN has resulted in an escalation
deployment of new services. The	novel centralized management paradigm introduced by	SDN has resulted in an escalation

Appendix F. Concordance lines of *IN DT JJ NN VBD TO* in CCE

Left context	KWIC	Right context
omponents [3], [4], [5], [7], [8], [9], [10]. The aim	of the present study was to	investigate whether identical simple movements
rators are used in the simulator. Thus, the goal	of the current study was to	determine skill transfer from simulator training to
for the introduced angular deviations. The goal	of the current study was to	extend this paradigm to investigate the role of c
ater willingness to return for treatment. The aim	of the current study was to	study both immediate and more long-term effect
nificantly reduced cost. Thus, a major objective	for the current review was to	quantitatively explore the effectiveness of VE an
nd had overlapping spectral envelopes. The aim	of the present study was to	assess the influence of sequential grouping cue
of VR for chronic pain syndromes. The purpose	of this present study was to	determine the impact of a brief VR session on th
ned for eliciting mindfulness. The main objective	of the present study was to	explore the feasibility, acceptability and the pote
differences in stress processing [9,10]. The aim	of the current study was to	explore the impact of individual differences in tra
sful as an established trauma film [47]. The aim	of the current study was to	test if individual differences in HA modulate affe
[41], and other emotional tests [42–44]. The aim	of the present study was to	replicate the Billington et al. [19] study in light of

Appendix G. Concordance lines of *DT JJ NN VBD TO VV* in CCE

Left context	KWIC	Right context
ponse could be changed after this practice.	A second aim was to assess	the extent to which self-reported measures
are used in the simulator. Thus, the goal of	the current study was to determine	skill transfer from simulator training to on-w
ces in stress processing [9,10]. The aim of	the current study was to explore	the impact of individual differences in trait a
introduced angular deviations. The goal of	the current study was to extend	this paradigm to investigate the role of cogn
lingness to return for treatment. The aim of	the current study was to study	both immediate and more long-term effects
an established trauma film [47]. The aim of	the current study was to test	if individual differences in HA modulate affe
overlapping spectral envelopes. The aim of	the present study was to assess	the influence of sequential grouping cues on
eliciting mindfulness. The main objective of	the present study was to explore	the feasibility, acceptability and the potential
ents [3], [4], [5], [7], [8], [9], [10]. The aim of	the present study was to investigate	whether identical simple movements produc

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d other emotional tests [42–44]. The aim of	the present study was to replicate	the Billington et al. [19] study in light of the
harder to elicit with a non-clinical sample).	The first aim was to gauge	VRE effectiveness of agent-based software
nd execution in new, untrained participants.	The main purpose was to establish	that the activation patterns from different su
E would elicit distress within each scenario.	The second aim was to determine	whether within-scenario and between-scena
e and absence of a metronome set to 4 Hz.	The secondary objective was to test	the reliability of these phenomena between
or chronic pain syndromes. The purpose of	this present study was to determine	the impact of a brief VR session on the exp

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